

OligoTox-Coagulopathy

Coagulation toxicity of oligonucleotide therapeutics · Narrative document · NIH/NCATS Oligonucleotide Toxicity Open Data Challenge, Phase 2

218

oligonucleotides

2,685

coagulation
measurements

1,039

per-position
modification records

100

sources, every
document held

112/112

QC checks pass

1. Executive summary

OligoTox-Coagulopathy pairs the design of 218 oligonucleotides with 2,685 measured coagulation outcomes drawn from 100 sources, every one of which is held in the release so that any row can be re-checked against the document it cites. 52 compounds carry a full per-nucleotide chemistry map, expanded into a 1,039-row position table; 104 of 218 (48%) carry a published sequence. Coagulopathy is the fifth endpoint on the Challenge's list of toxicities of interest, and before this release the public literature offered no structured, sequence-resolved account of it.

The endpoint is prolongation of clotting times, fibrinogen and factor disturbance, and the bleeding or thrombotic outcomes that follow from them. Platelet count alone is excluded: thrombocytopenia is a separate endpoint on the same list, and conflating the two is the commonest way this literature is misread.

The human evidence base

46 verified human interventional trials report a coagulation endpoint, 21 of them identified by a registry number. The register (`data/studies.csv`) holds 211 distinct study records, deduplicated across each study's registry entry, publication, regulatory assessment and label; 141 identified trials are registered and deliberately excluded, most because they report no coagulation endpoint. 42 pooled analyses are held in a separate table and never counted as trials, because their participants overlap the trials they pool. 424 of 749 clinical rows resolve to exactly one trial; the remaining 325 are left unresolved with their candidate count recorded rather than assigned to a guessed arm.

The human rows divide into 749 measurements in trial participants and 431 in human laboratory systems — primary blood or plasma, cells and tissue, and purified or recombinant human protein. These are not interchangeable evidence and the release keeps them apart.

Human-to-animal extrapolation

The Challenge names datasets able to extrapolate between in vitro human systems and animal data as of particular interest. **24 compounds carry the same readout category measured both in a human in vitro system and in an animal** — directly comparable pairs, listed in the `bridge` sheet. 29 compounds in total have human in vitro and animal data, and only 5 have participant and animal data. The three are reported separately because a shared assay is what makes a bridge usable; having some human row and some animal row does not.

Positive and negative controls

The dataset is not a list of toxic compounds. It spans the full range and carries controls at both ends, which is what makes it trainable:

- **Measured negatives — 640 rows.** An endpoint that was assessed and found unremarkable. These are distinguished, deliberately and throughout, from an endpoint that was never measured: the latter is `NOT_REPORTED` with the reason recorded, never `no_change`. Independent reviewers tested this distinction specifically and could not break it.
- **Vehicle, saline and untreated-plasma comparators** carried as the matched control on the row they belong to, in `control_value`, rather than as separate rows a user must find and join.

- **Severe positives — 28 rows at grade 3**, the highest severity the rubric assigns, plus 24 rows carrying a severity grade the source itself reported.
- **Pharmacological positive controls.** 1,790 rows measure compounds designed to alter coagulation — anti-factor-XI and anti-factor-XII antisense, prekallikrein and factor-VII programmes, anticoagulant aptamers, antithrombin-lowering siRNA. They are a mechanistically-explained positive class, and they are flagged so they are never mistaken for toxicity.

The single most important thing about this dataset. 1,790 of 2,685 rows are **on-target pharmacology, not toxicity**, because the compounds with published clotting numbers are largely the ones designed to change clotting. Two boolean columns keep the axes apart and **both may be true on one row** (183 rows are). A model trained across them without the flags learns that anticoagulants prolong aPTT — true, circular, and useless for safety prediction.

What the controls actually are, and where they are thin

Classified by the reference each row was compared against: vehicle or buffer 1,277 rows, placebo 223, pharmacological positive control 81 (enoxaparin, heparin, bivalirudin, warfarin, protamine), and **sequence-matched negative control 8 rows**. 5 compounds are included specifically as endpoint negative controls. The last figure is the one to read: a vehicle or a placebo does not separate a sequence effect from a chemistry or formulation effect, and only 8 rows in the release carry a scrambled or reverse-complement control. Anyone modelling sequence-dependent risk from this dataset is working largely without that control, and `control_class` makes that checkable rather than rhetorical.

Tested material: what is and is not characterised

Of the 38 compounds dosed in human participants, 14 have a printed sequence, 11 have position-resolved chemistry, 16 have identity confirmation recorded and 2 carry a purity value. 12 compounds are in a state worth naming precisely: the regulatory document *names* the purity test and withholds the numeric limit, which public assessment reports routinely do. That is recorded as a withheld limit, never as a value. Purity values belong to lots and are stored that way — tofersen reads 90% for one lot and 94% for another, so it carries no single purity figure. **No model can learn sequence-dependent toxicity from a compound whose sequence is not in the dataset**, and for the clinically dosed subset that is most of them.

What the citations do and do not establish

Every numeric value in the release was located in the document it cites: 2,019 checkable numbers, all found. That is a check against fabrication, not against misattribution — it proves the number is printed in the source, not that it belongs to the arm the row assigns it to. The 353 participant rows flagged as unintended toxicity were therefore audited separately for whether the cited locus names a table and whether the row's own quote identifies the arm: 291 supported, 59 weak, 3 unsupported. The per-row verdicts are in the `attribution_audit` sheet.

2. Main findings and conclusions

Finding 1 — the phosphorothioate class effect, quantified per compound for the first time

The safety literature states this effect at class level: prolongation of coagulation time at high plasma $C_{\max}$ of phosphorothioate-backbone oligonucleotides, independent of sequence and of hybridisation. This release expresses it as per-compound numbers. Across rows that are *not* on-target pharmacology, full-phosphorothioate compounds show a median aPTT of **1.35× the matched control** (n = 57).

The caveat is load-bearing and the figure is drawn to show it: **42 of those 57 rows come from a single source**. It is one well-controlled experiment, not a meta-analysis, and it should not be cited as though it were.

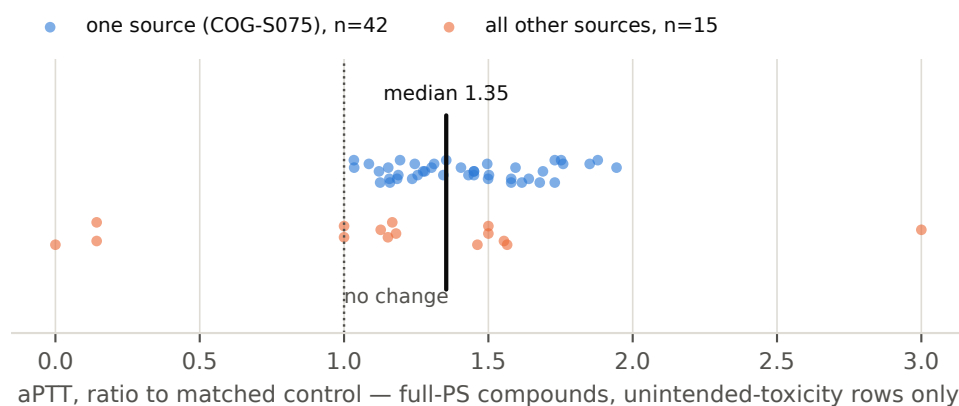


Figure 3. aPTT prolongation for full-phosphorothioate compounds, unintended-toxicity rows only. Points are split by source to make the concentration of evidence visible: 42 of 57 come from one study.

Finding 2 — a source whose own prose contradicts its own tables

US 9,061,044 states verbatim that "PT, aPTT and fibrinogen were not significantly altered in monkeys treated with ISIS oligonucleotides compared to the PBS control." Its Table 87 shows every treated group above control at every timepoint, in a clean compound rank order peaking at 4 h — 39.13 s against a 20.13 s control, a 1.94× prolongation. The dataset extracts the tables and carries the contradicting sentence verbatim on all 126 of that source's rows, so the disagreement travels with the data. Any pipeline that reads conclusions rather than tables records a false negative here.

Finding 3 — the standard in-vitro assay cannot do the job it is used for

One source demonstrates that in-vitro aPTT is saturated by phosphorothioate content and therefore cannot discriminate toxic from non-toxic compounds. Nulls from that assay are encoded as a *method limitation*, never as a safety finding. The opposite reading — the intuitive one — would teach a model that a saturated assay means a safe compound.

Finding 4 — human and animal evidence are separated, and rarely paired

1,183 measurements are made in human or human-derived systems and 1,476 in animal systems. Only **30 of 218 compounds carry both**, which bounds what any translation model built on this release can claim.

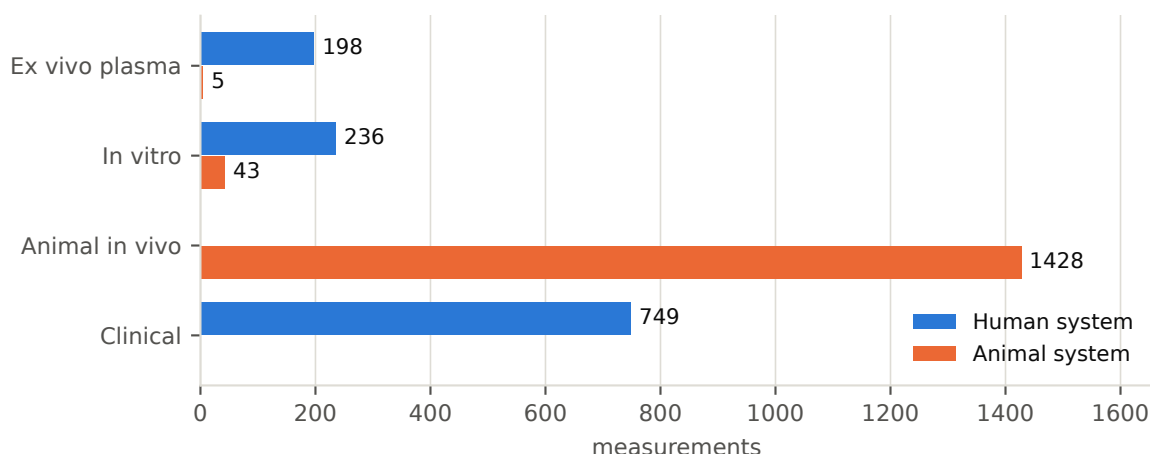


Figure 1. Human versus animal evidence by study design. The clinical and animal-in-vivo arms are disjoint by construction; the in-vitro and ex-vivo arms are where human systems and animal systems both appear.

3. How the data were produced

This is an **in-silico curation** of published data; no wet-lab experiment was performed. Sources were found by eight independent search axes — mechanism, clinical trials, nonclinical toxicology, siRNA and lipid nanoparticles, aptamers, regulatory labels, patents, and reviews used as a citation map —

followed by a targeted sweep for human data. Every candidate had to be *retrieved and quoted* before entering the work-list; nothing rests on a search-engine summary. That discipline closed the project's only previously recorded lead for this endpoint, which proved on retrieval not to report the values its title implied.

Extraction rules, enforced by the output contract: every measurement carries a verbatim quote and an exact locus; no value is ever read off a figure (614 rows are therefore qualitative rather than invented); per-position chemistry is transcribed from the source's own legend, never modelled. Retrieval used the Europe PMC REST service, DailyMed, the ClinicalTrials.gov API, EMA and FDA review documents, and the USPTO full-text endpoint. The pipeline is deterministic and committed: from a clean checkout, `build_dataset.py` → `validate_dataset.py` → `verify_against_sources.py` reproduces and re-checks the release with no network access.

Computational processing, and what it corrects

The build applies corrections that an adversarial verification pass identified, rather than leaving them as hand edits: a "relative" clotting time is a subtracted delta and must not be divided by its control; percent *inhibition* is not percent *of control*; a combination arm must be referenced to the partner-drug arm, not to the untreated cell; 120 pre-dose baselines are not effect measurements; and a source-stated measured null outranks a ratio a hair above 1.00. 17 rows carry a `co_administered_agent` and are not measurements of the oligonucleotide alone.

4. Indicators, predictors and their distributions

Response variables — how coagulation toxicity was measured

Axis	Distribution
Readout category	clotting time 1295 · factor activity 391 · bleeding outcome 306 · thrombotic outcome 252 · fibrinogen 145 · anticoagulant activity 97 · platelet coag crosstalk 96 · thrombin generation 56 · fibrinolysis marker 41 · blanket adverse event statement 3 · target transcript level 1 · complement marker 1 · infusion reaction 1
Readout (top)	aPTT 651 · PT 402 · fibrinogen 141 · FXI_activity 108 · antithrombin_activity 96 · thrombus_platelet_accumulation_fluorescence 65 · template_bleeding_time 58 · fibrin_formation_fluorescence 52
Study design	clinical 749 · animal in vivo 1,433 · in vitro 297 · ex vivo plasma 203
Species	human 1147 · monkey 818 · mouse 569 · rat 31 · pig 21 · minipig 17
System origin	human 1,183 · animal 1,476 · not determined 26
Human system subtype	participants 749 · human laboratory 431
Kind of observation	intended pharmacodynamic 971 · measured negative 640 · unintended lab disturbance 408 · outcome not attributed 292 · adverse outcome source attributed 160 · baseline reference 120 · unresolved observation 77 · unattributed lab change 17
Who graded it	ungraded 1748 · curator derived research score 913 · source reported 19 · both source reported and curator derived 5
Control compared against	vehicle or buffer 1277 · other described control 727 · untreated or predose 314 · placebo 223 · pharmacological positive control 81 · active comparator 28 · no control described 27 · sequence control 8

Grades are ordinal 0–3, assigned **mechanically** from a control-referenced ratio by published **CTCAE v5.0** cut-offs, and only for the readouts CTCAE defines. 1,767 rows (66%) are deliberately **left ungraded**, each stating why, rather than graded by a threshold invented here.

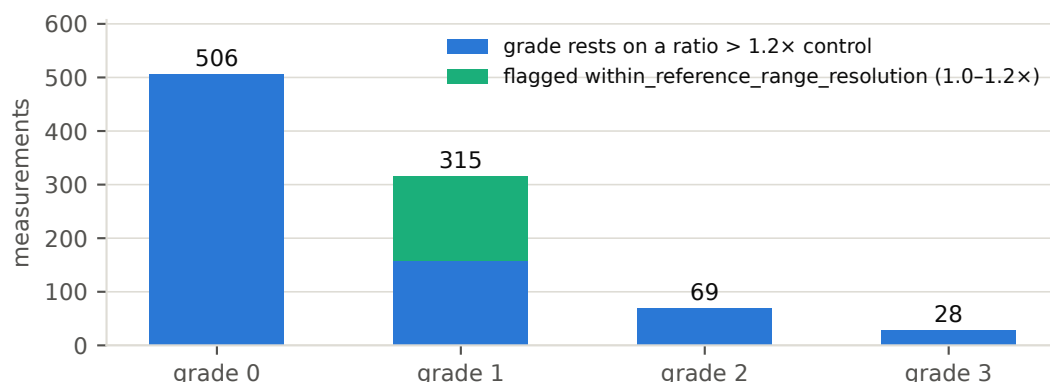


Figure 4. Grade distribution. The upper segment marks grades resting on a ratio between 1.0 and 1.2× control, which normal variation cannot be excluded from.

Read the grade column with its caveat. CTCAE grades against the upper limit of normal; these sources publish a control mean. A ratio a few percent above 1.00 is therefore not evidence of a real prolongation. 157 graded rows carry `grade_caveat = within_reference_range_resolution`. Filter on it before treating grade 1 as a finding.

Predictor variables — the distribution across tested oligos

Variable	Distribution across 218 compounds
Class	ASO gapmer 61 · aptamer 60 · other 36 · GalNAc siRNA 28 · siRNA 13 · tcDNA ASO 7 · polydisperse ssDNA 4
Backbone	NOT REPORTED 100 · full PS 59 · mixed PO PS 36 · full PO 14 · PMO neutral 4
Sequence published	104 / 218
Position-resolved chemistry	52 / 218 (1,039 position records)
Same readout in a human in vitro system <i>and</i> an animal	24 / 218
Human in vitro <i>and</i> animal data	29 / 218
Participant <i>and</i> animal data	5 / 218
Purity value / purity method named / identity confirmed	2 / 11 / 16 of the 38 dosed in participants

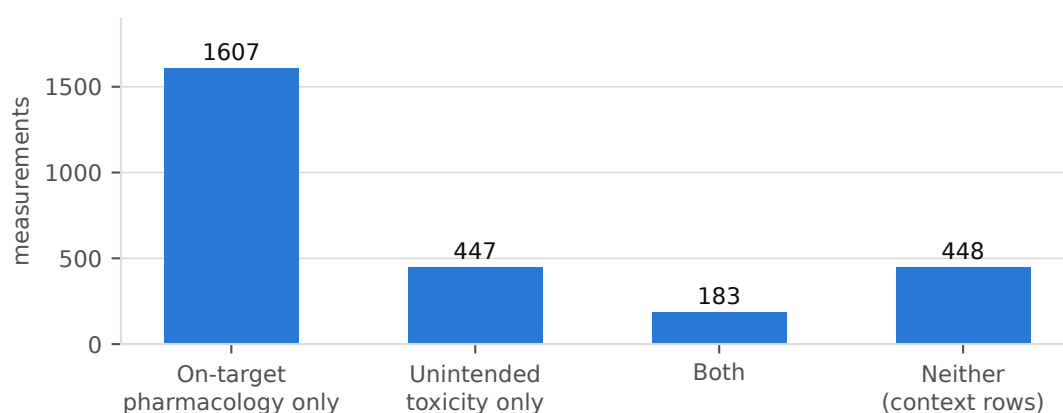


Figure 2. The two axes. On-target pharmacology dominates the row count, which is why the flags exist.

5. The gap this addresses

Coagulopathy is named in the Challenge brief and is, in the published record, described almost entirely at class level. The characterisation the field works from — that phosphorothioate backbones prolong clotting time at high $C_{\max}$, independent of sequence — appears in review and methods literature as a sentence, naming no compound, no dose and no value. What did not exist was a structured table pairing an identified oligonucleotide, its sequence and its per-position chemistry, with a measured coagulation outcome and a citable locus. This release is that table.

It also fills a narrower gap the field has been explicit about: the separation of *on-target anticoagulant pharmacology* from *unintended coagulation toxicity*. Those two live side by side in the literature — often in the same paper — and are routinely pooled. Here they are two columns.

Finally, the release distinguishes human from animal evidence at row level, including for purified-protein systems, where the human/animal question is invisible in a species field. 1,183 rows (44%) are human or human-derived, 434 of them from human in-vitro and ex-vivo systems — the category the Challenge identifies as of particular interest.

6. Using this to build a predictive model

The four-table design exposes sequence, per-position chemistry and design predictors against graded, per-condition outcomes, with no join required beyond `oligo_id`. A model builder should:

- **Train on the unintended-toxicity class** (447 rows, plus the 183 rows where an on-target compound produced harm), using the 1,607 on-target rows as a mechanistically-explained positive class rather than as toxicity labels.
- **Use the 640 measured nulls as negatives**, and *not* the `NOT_REPORTED` rows, which encode absence of measurement, not absence of effect. The distinction is the difference between a safety model and a model of what has been studied.
- **Filter `grade_caveat`** before treating grade 1 as signal, and prefer `ratio_to_control` — a continuous target — over the ordinal grade.
- **Respect `species_class`**. Train and validate within a system class, or model the human/animal gap explicitly using the 30 compounds that carry both.
- **Exclude `co_administered_agent` rows** from single-agent models.

The most defensible near-term target is not a grade classifier but a regression on aPTT ratio from backbone composition and phosphorothioate count, trained on the unintended class — the one relationship this release now supports with per-compound numbers.

7. Limitations, stated plainly

- **Grades are provisional and mechanical.** No subject-matter expert has reviewed them.
- **The class-effect quantification rests on one source** (42 of 57 rows).
- **No clinical compound has a published sequence** in the sources used, so sequence-to-phenotype modelling is restricted to patent and preclinical compounds.
- **26 rows have an undetermined system origin** — the source never states whether the proteins used were human. They are marked, not guessed.
- **Prothrombotic rows are dominated by one compound family**, so "hypercoagulability" risks being learned as one drug.
- **6 rows are scope-adjacent** — retained as context but marked, because they are not coagulation readouts.
- **Open defects are enumerated**, not summarised away, in the endpoint dossier.

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